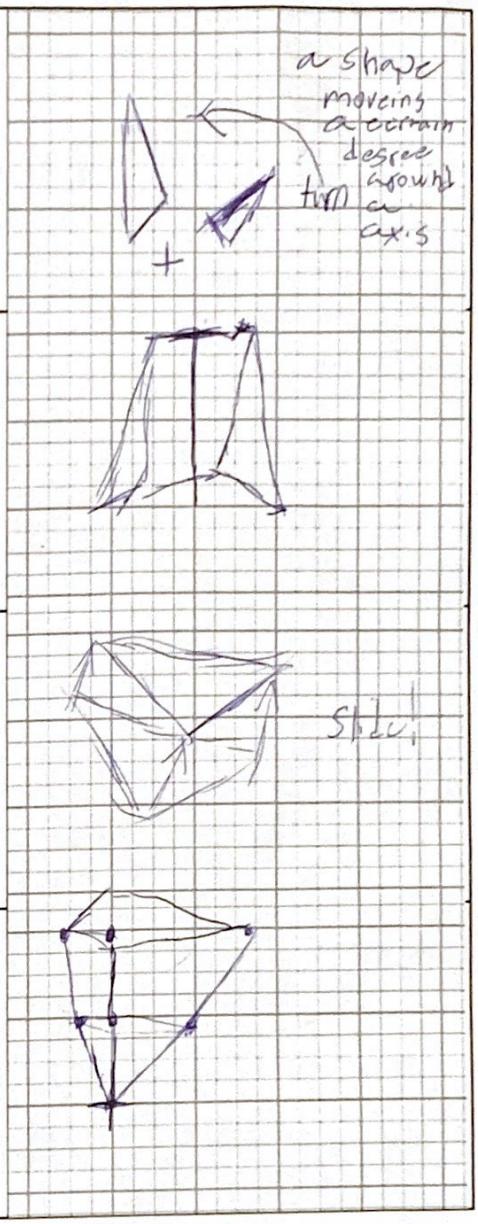
Transformational Geometry

Class: Transition Year **Date:** 14/09/2020

Transformational geometry involves the movement of objects and points.

Use www.mathsisfun.com/geometry/transformations.html to help you write your own definition of the **4 Transformations.**

1. Rotation:	
2. Reflection:	
3. Translation:	
4. Resizing:	

What is different about Resizing compared to Rotating, Reflecting and Translating?

Log on to mangahigh.com/login to practice your Transformation skills.

User ID: yourfullname Password: yourfirstname123 School ID: 470059

Transformational Geometry

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Transformational geometry involves the movement of objects and points.

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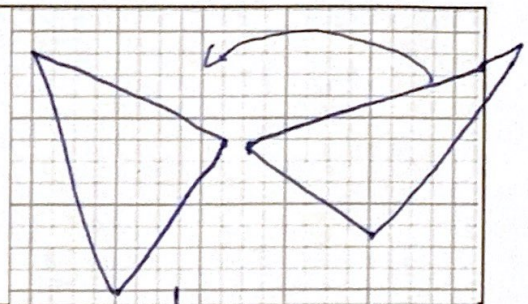
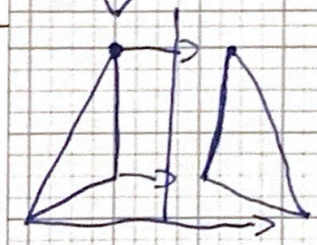
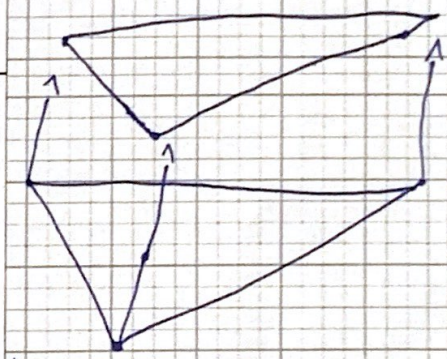
1. Rotation:



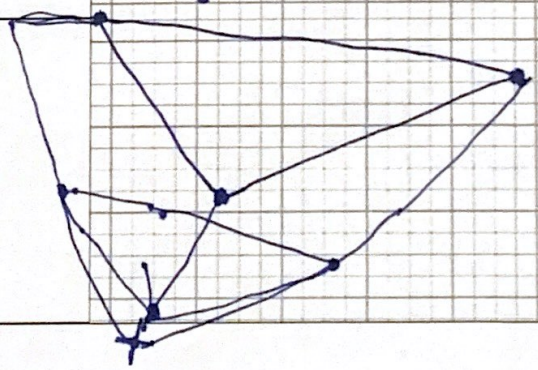
2. Reflection:



3. Translation:



4. Resizing:



What is different about Resizing compared to Rotating, Reflecting and Translating?

_____

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Transformational Geometry

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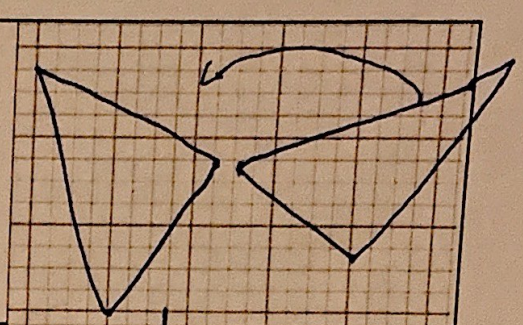
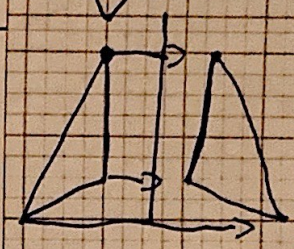
Transformational geometry involves the movement of objects and points.

Use www.mathsisfun.com/geometry/transformations.html to help you write your own definition of the **4 Transformations.**

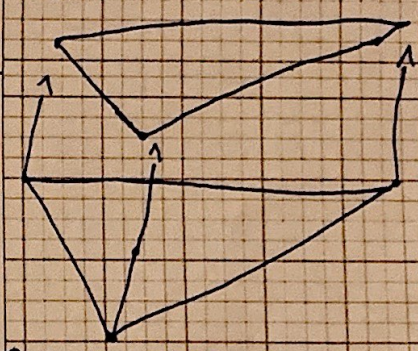
1. Rotation:



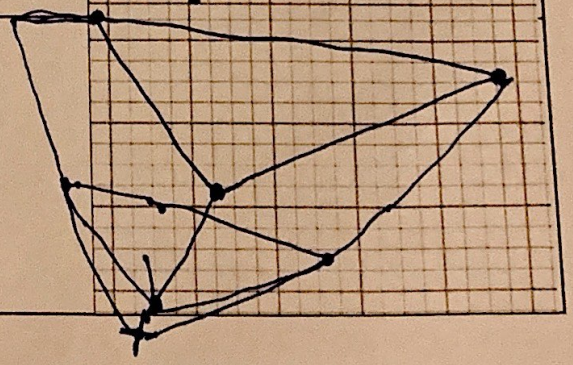
2. Reflection:



3. Translation:



4. Resizing:



What is different about Resizing compared to Rotating, Reflecting and Translating?

_____

Log on to mangahigh.com/login to practice your Transformation skills.

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..ansformational Geometry

Class: Transition Year **Date:** *14/09/2020*

Transformational geometry involves the movement of objects and points.

Use www.mathsisfun.com/geometry/transformations.html to help you write your own definition of the **4 Transformations.**

1. Rotation:	Rotation means turning around a center.
2. Reflection:	You get a reflection in an image when every point is the same distance from the 'central' or 'mirror' line.
3. Translation:	Translation in geometry simply means moving.
4. Resizing:	When you resize a shape it gets larger or smaller.

What is different about Resizing compared to Rotating, Reflecting and Translating?

Log on to mangahigh.com/login to practice your Transformation skills.

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Equivalent Fractions

1. On your fraction wall, colour $\frac{2}{4}$ in red
   What fraction is this the same as? $\frac{3}{6}$

2. On your fraction wall, colour $\frac{2}{8}$ in blue
   What fraction is this the same as? $\frac{3}{12}$

3. On your fraction wall, colour $\frac{2}{6}$ in purple
   What fraction is this the same as? $\frac{4}{12}$

4. On your fraction wall, colour $\frac{2}{12}$ in yellow
   What fraction is this the same as? $\frac{1}{6}$

5. On your fraction wall, colour in $\frac{2}{10}$ in green
   What fraction is this the same as? $\frac{1}{5}$

6. Circle which of the fractions are equivalent to $\frac{1}{2}$

   $\frac{3}{6}$ $\frac{5}{8}$ $\frac{3}{5}$ $\frac{6}{12}$ $\frac{5}{10}$

7. Circle which of the fractions are equivalent to $\frac{1}{3}$

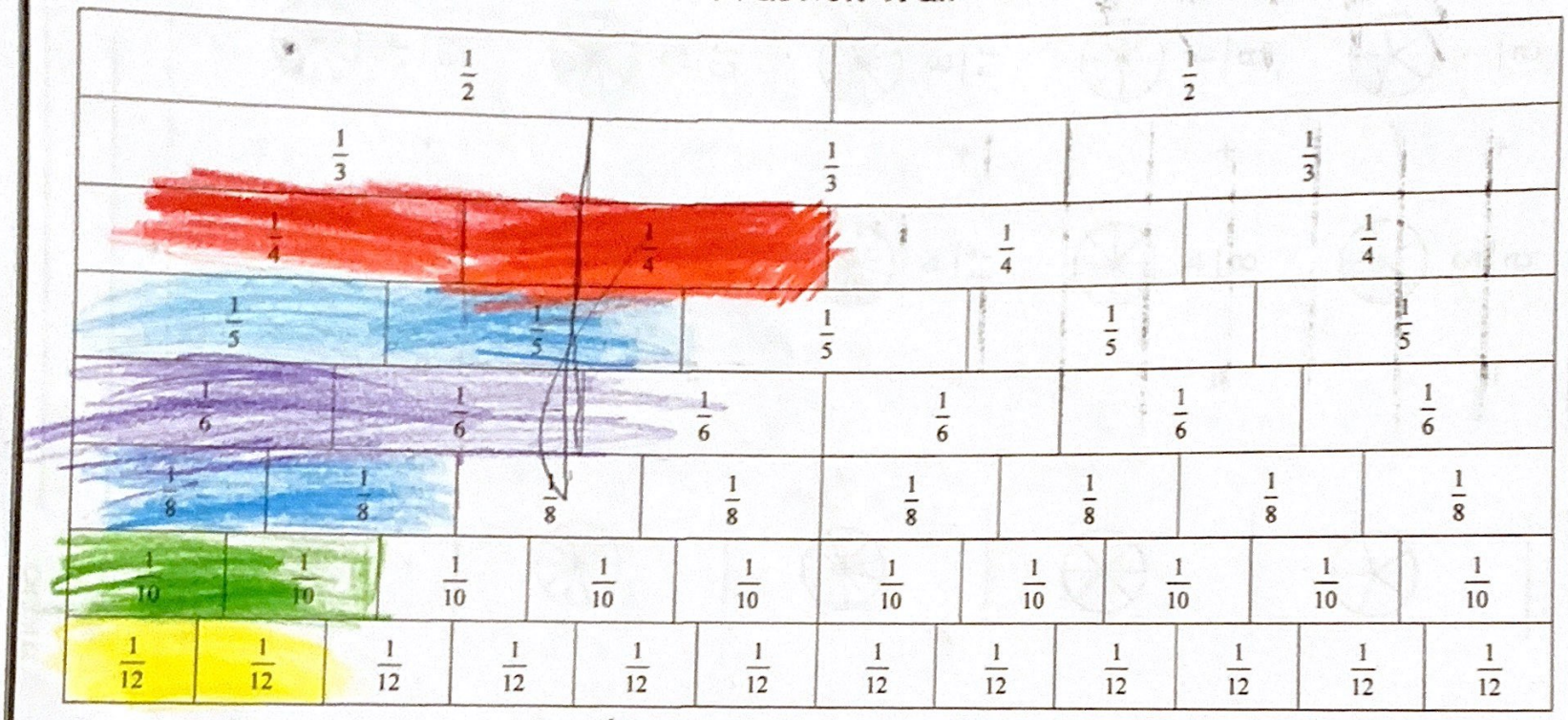
   $\frac{1}{2}$ $\frac{3}{8}$ $\frac{2}{6}$ $\frac{4}{12}$ $\frac{3}{10}$

8. Circle which of the fractions are equivalent to $\frac{1}{4}$

   $\frac{2}{8}$ $\frac{2}{3}$ $\frac{2}{6}$ $\frac{3}{12}$ $\frac{2}{10}$

5

Fraction Wall



5

Fraction Wall

$\frac{1}{2}$						$\frac{1}{2}$					
$\frac{1}{3}$			$\frac{1}{3}$			$\frac{1}{3}$					
$\frac{1}{4}$			$\frac{1}{4}$			$\frac{1}{4}$			$\frac{1}{4}$		
$\frac{1}{5}$		$\frac{1}{5}$		$\frac{1}{5}$		$\frac{1}{5}$		$\frac{1}{5}$			
$\frac{1}{6}$			$\frac{1}{6}$		$\frac{1}{6}$		$\frac{1}{6}$		$\frac{1}{6}$		
$\frac{1}{8}$				$\frac{1}{8}$		$\frac{1}{8}$		$\frac{1}{8}$		$\frac{1}{8}$	
$\frac{1}{10}$					$\frac{1}{10}$		$\frac{1}{10}$		$\frac{1}{10}$		
$\frac{1}{12}$						$\frac{1}{12}$					

Nam

ntegers

Date: 7/10/20

1. Solve these sums involving *addition* and *subtraction* of integers:

3 + 7 10	12 – 6 6	– 8 + 4 -4	–10 – 3 -13
6 + 7 – 5 8	20 – 8 – 5 7	? 6 – 2 – 9 +17	7 + 9 – 3 13

2. Solve these sums involving *multiplication* of integers:

6 x 4 24	3 x 10 30	–2 x 5 -10	4 x (– 4) -16
(–7) x (–2) -9 14	(– 8) x 3 -24 -16	11 x (–3) -33	(–10) x (–5) +50

3. Solve these sums involving *division* of integers:

$\frac{6}{2}$	$\frac{-8}{4}$	$\frac{-9}{3}$	$\frac{-10}{-2}$
3	−2	−3	+20 +5 5
$\frac{20}{-5}$	$\frac{18}{9}$	$\frac{30}{-10}$	$\frac{-44}{-11}$
−4	−2	−3	+4

4. Can you solve these problems involving *addition, subtraction, multiplication, and division?*

8 − 3 + 4 x (2)	10 + 2 x (− 3)
~~8−3+4x(2)~~ 8−3+ 4x2 8 13	10x2x −3 10−6 9 4
15 + 5 − $\frac{6}{3}$	$\frac{12}{4}$ + 20 − (5 x 2)
15+5− 20−2=18 20	3+20 10−2 23−3 10−2 20−6 +1

3+20−10
3+10 =13

Name:

Integers

Date: 7/10/20

1. Solve these sums involving *addition* and *subtraction* of integers:

3 + 7 3 + 7=10	12 – 6 12 – 6 = 6	– 8 + 4 – 8 + 4 = -12 X -8 +4 = -4	–10 – 3 -10 – 3 = -7 -10-3=-13
6 + 7 – 5 6 + 7 - 5 = +8	20 – 8 – 5 20 - 8 - 5 = +7	6 – 2 – 9 6 - 2 - 9 = -5	7 + 9 – 3 7 + 9 - 3 = 13

2. Solve these sums involving *multiplication* of integers:

6 x 4 6 x 4 = 24	3 x 10 3 x 10 = 30	–2 x 5 -2 x 5 = -10	4 x (– 4) 4 x -4 = -16
(–7) x (–2) -7 x - 2 = +14	(– 8) x 3 -8 x 3 = -24	11 x (–3) 11 x - 3 = -33	(–10) x (–5) -10 x -5 = +50

3. Solve these sums involving *division* of integers:

$\frac{6}{2}$ $\frac{6}{2} = 3$	$\frac{-8}{4}$ $\frac{-8}{4} = -2$	$\frac{-9}{3}$ $\frac{-9}{3} = -3$	$\frac{-10}{-2}$ $\frac{-10}{-2} = +5$
$\frac{20}{-5}$ $\frac{20}{-5} = -4$	$\frac{18}{9}$ $\frac{18}{9} = 2$	$\frac{30}{-10}$ $\frac{30}{-10} = -3$	$\frac{-44}{-11}$ $\frac{-44}{-11} = +4$

4. Can you solve these problems involving *addition, subtraction, multiplication, and division?*

$8 - 3 + 4 \times (2)$ $8 - 3 + 4 \times 2 = 13$ $8 - 3 = + 8 =$	$10 + 2 \times (-3)$ $10 + 2 \times -3$ $2 \times -3 = -6$ $10 + -6 = 4$
$15 + 5 - \frac{6}{3}$ $15 + 5 = 20 - 2 = 18$ $\frac{6}{3} = 2$	$\frac{12}{4} + 20 - (5 \times 2)$ $23 + 10 = 33$ $\frac{12}{4} = 3$ $5 \times 2 = 10$

Name: [redacted] **Integers** Date: _____

1. Solve these sums involving *addition* and *subtraction* of integers:

3 + 7 10	12 – 6 6	– 8 + 4 -4	–10 – 3 -13
6 + 7 – 5 13 – 5 8	20 – 8 – 5 12 – 5 7	6 – 2 – 9 4 – 9 -5	7 + 9 – 3 16 – 3 13

2. Solve these sums involving *multiplication* of integers:

6 x 4 24	3 x 10 30	–2 x 5 -10	4 x (– 4) -16
(–7) x (–2) +14	(– 8) x 3 -24	11 x (–3) 11 +11 33	(–10) x (–5) +50

3. Solve these sums involving *division* of integers:

$\frac{6}{2}$ 3	$\frac{-8}{4}$ −2	$\frac{-9}{3}$ −3	$\frac{-10}{-2}$ +5
$\frac{20}{-5}$ −5	$\frac{18}{9}$ +2	$\frac{30}{-10}$ −3	$\frac{-44}{-11}$ +4

4. Can you solve these problems involving *addition, subtraction, multiplication, and division?*

8 − 3 + 4 x (2) 8−3+8 5+8 13	10 + 2 x (−3) 10+6 4
15 + 5 − $\frac{6}{3}$ 20−2 18	$\frac{12}{4}$ + 20 − (5 x 2) 3+20−10 23−10 13

100%

Percentages

Date: 3.11.2020

1. Using your calculator, find these percentages:

(a) 20% of 5	(b) 10% of 120	(c) 30% of 900	(d) $45\frac{1}{2}$% of 600
20% x 5 = 1	10% x 120 = 12	30% x 900 = 270	600 x $45\frac{1}{2}$% = $27\frac{2}{3}$

2. **Increase** or **decrease** these numbers:

(a) Increase 72 by 25%	(b) Increase 300 by 60%	(c) Increase 1000 by $55\frac{1}{2}$%	(d) Increase 450 by $33\frac{1}{3}$%
72 x 25% = 18 72 + 18 = 90	300 x 60% = 180 300 + 180 = 408	1000 x $55\frac{1}{2}$% = 555 1000 + 555 = 1555	450 x $33\frac{1}{3}$% = ~~15.75~~ 150 450 + ~~15.75~~ 150 = 600 ~~465.75~~

(e) Decrease 30 by 10%	(f) Decrease 150g by 30%	(g) Decrease €1200 by $32\frac{1}{2}$%	(h) Decrease 3000g by 36.5%
30 x 10% = 3 30 − 3 = 27	150 x 30% = 45 150 − 45 = 105	€1200 x $32\frac{1}{2}$% = €390 1200 + 390 = 810	3000g x 36.5% = 1095g 3000 − 1095 =

3. Find the selling price of a piece of furniture marked €850 + **VAT** at 20%

Vat €€850 X 20·1 = 170
total €850 + 170 = 1020

4. Find the selling price of an iPhone marked €700 + **VAT** at $22\frac{1}{2}$%

Vat €700 x $22\frac{1}{2}$·1 = €157·5
total €700 + 157.5 = €857·5

5. A Spotify subscription costs €12 per month, + **VAT** at 30%.
What is the total cost of Spotify for **6 months**?

Vat € 12 X 30·1 = 3·6
total 12 + 3·6 = 15·6
15·6 x 6 = 93·6

6. A shop is having a **SALE** for Black Friday, find the sale price of these items:
(a) Selfie Stick is normally €20, now with a 10% **discount**:

€20 x 10·1 = 2
20 + 2 = ~~€8~~ 18

(b) XBOX is normally €489, now with a 15% **discount**:

489 x 15·1 = 73·35
489 + 73·35 = 415·65

88%

Percentages

Date: 24/11/20

1. Using your calculator, find these percentages:

(a) 20% of 5	(b) 10% of 120	(c) 30% of 900	(d) $45\frac{1}{2}$% of 600
1.25	12	270	273

2. **Increase** or **decrease** these numbers:

(a) Increase 72 by 25%	(b) Increase 300 by 60%	(c) Increase 1000 by $55\frac{1}{2}$%	(d) Increase 450 by $33\frac{1}{3}$%
18 72 +18 90	+180+ 300 +180 480	555 1000 +555 1555	150 450 +150 600

(e) Decrease 30 by 10%	(f) Decrease 150g by 30%	(g) Decrease €1200 by $32\frac{1}{2}$%	(h) Decrease 3000g by 36.5%
31 21 30 −3 27	Ans 45 150 −45 105	390 1200 −390 810	1095 3000 −1095 1905

~~Common test~~ test

3. Find the selling price of a piece of furniture marked €850 + **VAT** at 20%

170

$\times \frac{170}{20}$ Vat 170

850
+170
€1020

4. Find the selling price of an iPhone marked €700 + **VAT** at $22\frac{1}{2}$%

Vat 157.5

700
+157.5
€857.5

5. A Spotify subscription costs €12 per month, + **VAT** at 30%.
What is the total cost of Spotify for **6 months**?

21.6
+7.2
28.83

Vat 3.6

12
× 6
72

21.6

3.6
× 6
21.6

72
+ 3.6
+ 28.8
93.

6. A shop is having a **SALE** for Black Friday, find the sale price of these items:
(a) Selfie Stick is normally €20, now with a 10% **discount**:

Vat = 2 ~~€20 × 10% =~~

Discount = €2

20
− 2
€18

10
− 2
€8

(b) XBOX is normally €489, now with a 15% **discount**:

15% = €73.35

73.35
− 15
1100.25

73.35
− 489
€810

100%

Percentages

Date: 24/11/20

1. Using your calculator, find these percentages:

(a) 20% of 5	(b) 10% of 120	(c) 30% of 900	(d) $45\frac{1}{2}$% of 600
= 1	= 12	= 270	= 273

2. **Increase** or **decrease** these numbers:

(a) Increase 72 by 25%	(b) Increase 300 by 60%	(c) Increase 1000 by $55\frac{1}{2}$%	(d) Increase 450 by $33\frac{1}{3}$%
18+72 = 90	180+300 = 480	555+1000 =1555	150×450 600

(e) Decrease 30 by 10%	(f) Decrease 150g by 30%	(g) Decrease €1200 by $32\frac{1}{2}$%	(h)Decrease 3000g by 36.5%
3+30 = 27 30−3 =	~~45+150~~ = 150+45 = 105	~~390+1200~~ 1200+390 810	~~1095−3000~~ 0 = 1905 3000+1095 = ~~4096~~ 1095

3000
−1905

3. Find the selling price of a piece of furniture marked €850 + **VAT** at 20%

Vat €170 Price 1020

4. Find the selling price of an iPhone marked €700 + **VAT** at $22\frac{1}{2}$%

vat €157.50 Price €857.50

5. A Spotify subscription costs €12 per month, + **VAT** at 30%.
What is the total cost of Spotify for **6 months**?

Vat: €3.60 1month Price €15.60

6 0 month

= €93.60

6. A shop is having a **SALE** for Black Friday, find the sale price of these items:
(a) Selfie Stick is normally €20, now with a 10% **discount**:

Discont €2.00 = Total = €18.00

(b) XBOX is normally €489, now with a 15% **discount**:

€73.35

= 415.65

100%

Percentages

Date: 24/11/20

1. Using your calculator, find these percentages:

(a) 20% of 5	(b) 10% of 120	(c) 30% of 900	(d) $45\frac{1}{2}$% of 600
20% × 5 = 1	120 × 10% = 12	900 × 30% = 270	600 × 45½ = 273

2. **Increase** or **decrease** these numbers:

(a) Increase 72 by 25%	(b) Increase 300 by 60%	(c) Increase 1000 by $55\frac{1}{2}$%	(d) Increase 450 by $33\frac{1}{3}$%
72 × 25% = 18 72 +18 90	300 × 60% = 180 300 + 180 480	1000 × 55½% = 555 1000 + 555 1555	450 × 33⅓% 150 450 + 150 600
(e) Decrease 30 by 10%	**(f)** Decrease 150g by 30%	**(g)** Decrease €1200 by $32\frac{1}{2}$%	**(h)** Decrease 3000g by 36.5%
30 × 10% = 3 30 - 3 27	150 × 30% 45 150 - 45 105	1200 × 32½% = 390 1200 - 390 810	3000g × 36.5% = 1095 3000 +1095 1905

3. Find the selling price of a piece of furniture marked €850 + **VAT** at 20%

vat = 170
Total = 1020

4. Find the selling price of an iPhone marked €700 + **VAT** at $22\frac{1}{2}$% ?

~~vat = 157.5~~
~~Total = 857.5~~

vat = 157.5
Total = 857.5

5. A Spotify subscription costs €12 per month, + **VAT** at 30%.
What is the total cost of Spotify for **6 months**?

vat = 3.6
~~total = 9.6~~ ~~Total = 72~~
Total one month 15.6
6 months = 93.6

6. A shop is having a **SALE** for Black Friday, find the sale price of these items:
(a) Selfie Stick is normally €20, now with a 10% **discount**:

Discount = 2
Total = 18

(b) XBOX is normally €489, now with a 15% **discount**:

vat = 73.35
Total 562.35

15.6

100%

Percentages

Date: 24/11/20

1. Using your calculator, find these percentages:

(a) 20% of 5	(b) 10% of 120	(c) 30% of 900	(d) $45\frac{1}{2}$% of 600
1	12	270	273

2. **Increase** or **decrease** these numbers:

(a) Increase 72 by 25%	(b) Increase 300 by 60%	(c) Increase 1000 by $55\frac{1}{2}$%	(d) Increase 450 by $33\frac{1}{3}$%
90	~~180~~ 480	~~555~~ 1555	$\frac{603}{4}$ 1,90

(e) Decrease 30 by 10%	(f) Decrease 150g by 30%	(g) Decrease €1200 by $32\frac{1}{2}$%	(h) Decrease 3000g by 36.5%
3 27	45 105	390 810	3000 −1095 1905 3000 −1905

clear

3. Find the selling price of a piece of furniture marked €850 + **VAT** at 20%

VAT = €170
Total price = €1020

4. Find the selling price of an iPhone marked €700 + **VAT** at $22\frac{1}{2}$%

~~VAT 15% = 8.57.5~~

Vat 157.5 = 857.5

5. A Spotify subscription costs €12 per month, + **VAT** at 30%.
What is the total cost of Spotify for **6 months**?

Vat = 3.6 = 15.6

6 months = 93.6

6. A shop is having a **SALE** for Black Friday, find the sale price of these items:
(a) Selfie Stick is normally €20, now with a 10% **discount**:

Vat 2 = 22

−20 = 18

(b) XBOX is normally €489, now with a 15% **discount**:

15% = 73.35 415.65

Cereal	M	P	C	B
No	20	12	25	15
	100	60	125	75

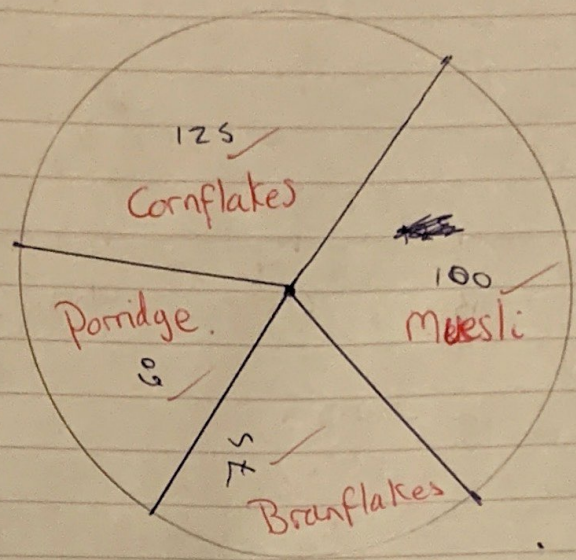
= 72

$\frac{20}{72} \times 360 = 100$

$\frac{12}{72} \times 360 = 60$

$\frac{25}{72} \times 360 = 125$

$\frac{15}{72} \times 360 = 75$



Label pie chart with names of data

$\frac{3}{4}$

Stem	leaf
4	7 7
5	0 1 9
6	1 2 3 5 7
7	3 6 7 8
8	6

mode = 47

median = 62 × 63

Range
47 - 86 = 39.

$\frac{3}{5}$

Key 7|3 = 73.